Expressions Workshop: Distribute, Combine, Factor

Grade 6–7 Expressions and Equations

Focus: Distributing over a sum, combining like terms, distributing then combining, and factoring out a common factor.

Every problem comes with a representation — an area model, a tape diagram, or a sorting table — that stands for the expression printed with it. Fill the representation in first, then write the simplified expression. Two expressions are *equivalent* when they give the same value for every number you could substitute, so a simplified expression must keep every term you started with.

1. A rectangle is 4 metres tall. Its base is cut into a piece x metres long and a piece 7 metres long, so its total area is $4(x+7)$ square metres. Write each piece's area in the model, then write the total area as an expanded expression (no parentheses).

	x	7
4		

Answer: ________________

2. A crew paints 5 identical rooms. Each room takes $(3x-2)$ minutes, where x is the number of minutes per wall. The tape diagram shows the whole job. Write the total time as an expanded expression in minutes.

$3x-2$	$3x-2$	$3x-2$	$3x-2$	$3x-2$

Answer: ________________

3. Two boxes of tiles are combined: $7x + 3$ tiles in the first and $2x + 8$ tiles in the second, where x is the number of tiles in one full row. Sort the terms into the table, then write the simplified expression for $7x + 3 + 2x + 8$.

terms with x	**number terms**

Answer: ________________

4. Tia has 8 packs of markers with m markers in each pack, plus 5 loose markers. She gives away 3 whole packs and then receives 2 more loose markers. That is $8m + 5 - 3m + 2$ markers. Sort the terms and simplify.

terms with m	**number terms**

Answer: ________________

5. A walking path has 3 curved sections, each $(x + 4)$ metres long, followed by a straight run of $2x$ metres. The total length in metres is $3(x + 4) + 2x$. Fill in the model, then write the simplified total length.

$x+4$	$x+4$	$x+4$	$2x$

Answer: ________________

6. Now run an area model *backwards*. A rectangle is split into two pieces of area $6x$ square units and 15 square units, so its total area is $6x + 15$ square units. Both pieces share the same height. Find the greatest common factor of $6x$ and 15, write it as the height, and write the total area as a product.

	$6x$	15

Answer: ____________________

7. A club orders 4 boxes holding $(2x+3)$ badges each, then returns 3 boxes holding $(x-2)$ badges each. The badges kept are $4(2x + 3) - 3(x - 2)$. Distribute both products in the table — watch the subtraction — then combine like terms.

add 4 groups of $(2x + 3)$		
subtract 3 groups of $(x - 2)$		

Answer: ____________________

8. A banner starts as a rectangle of area $12n$ square inches. A strip of area 18 square inches is cut off, leaving $12n - 18$ square inches. The rectangle and the strip have the same height. Use the backwards area model to write $12n - 18$ as a product with the greatest common factor outside.

	$12n$	-18

Answer: __________________

9. A tutoring centre charges 5 dollars per session for each of $(x + 3)$ students and adds a materials fee of $4x$ dollars, so the cost in dollars is $5(x + 3) + 4x$.

Given	**Cost in dollars**
5 dollars per session, $(x + 3)$ students	$5(x + 3)$
materials fee	$4x$

(a) Simplify the total cost expression.

Answer: __________________

(b) There are 7 students in the group, so $x = 7$. Find the total cost in dollars.

Answer: __________________

10. Challenge — find and fix the error. Jamal simplified $3(2x - 5)$ like this:

Jamal's work: $3(2x - 5) = 6x - 5$

Say which number he failed to multiply and why the distributive property requires it, then write the correct expanded expression.

Answer: ____________________